Project Title

Symptom-to-Diagnosis Matcher

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Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Symptom-to-Diagnosis Matcher is a clinical decision-support system designed to convert free-text patient symptoms into ranked lists of probable medical conditions. The proposed engine uses natural language processing (NLP) to normalize symptom descriptions by handling spelling errors, abbreviations, synonyms, and variations in expression. Extracted symptoms are matched against a structured medical knowledge base containing diseases, associated symptoms, and diagnostic test information. A transparent weighted scoring mechanism evaluates symptom relevance, frequency, and importance to generate confidence scores for each candidate diagnosis. The system also recommends a minimal set of discriminating diagnostic tests that can effectively distinguish among the top-ranked conditions, reducing unnecessary investigations and improving efficiency. By combining NLP, knowledge-based reasoning, interpretable scoring, and intelligent test recommendation, the proposed approach supports accurate preliminary diagnosis, assists healthcare professionals in clinical triage, and enhances decision making in telemedicine, primary healthcare screening, and medical education environments.

Signature of the Guide